Adaptive-RAG: Choosing a Retrieval Strategy by Question Complexity

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Based on: S. Jeong, J. Baek, S. Cho, S. J. Hwang, J. C. Park. "Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity". NAACL 2024, pp. 7036–7050

Why Retrieval?

LLM knowledge is stored in the model parameters

It can be incomplete or out of date, so models give wrong facts

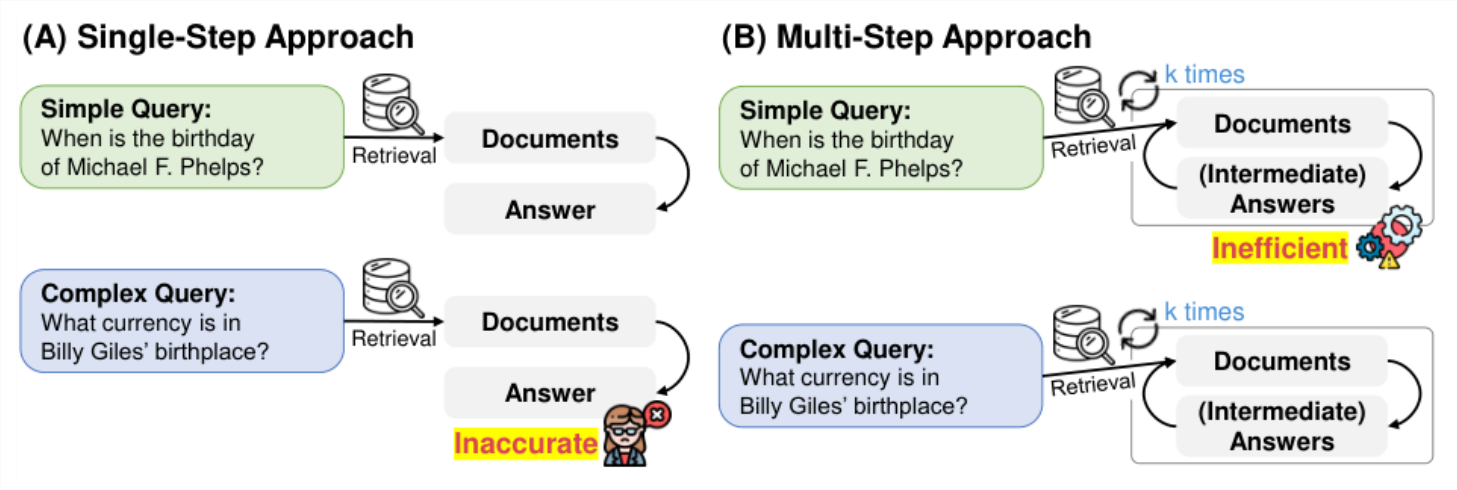
Retrieval: search documents (e.g. Wikipedia) before answering

Every search step costs time and computation

Outline

1. The problem
2. The method
3. Experiments and main results
4. What the results mean

The Problem: Questions Differ in Complexity

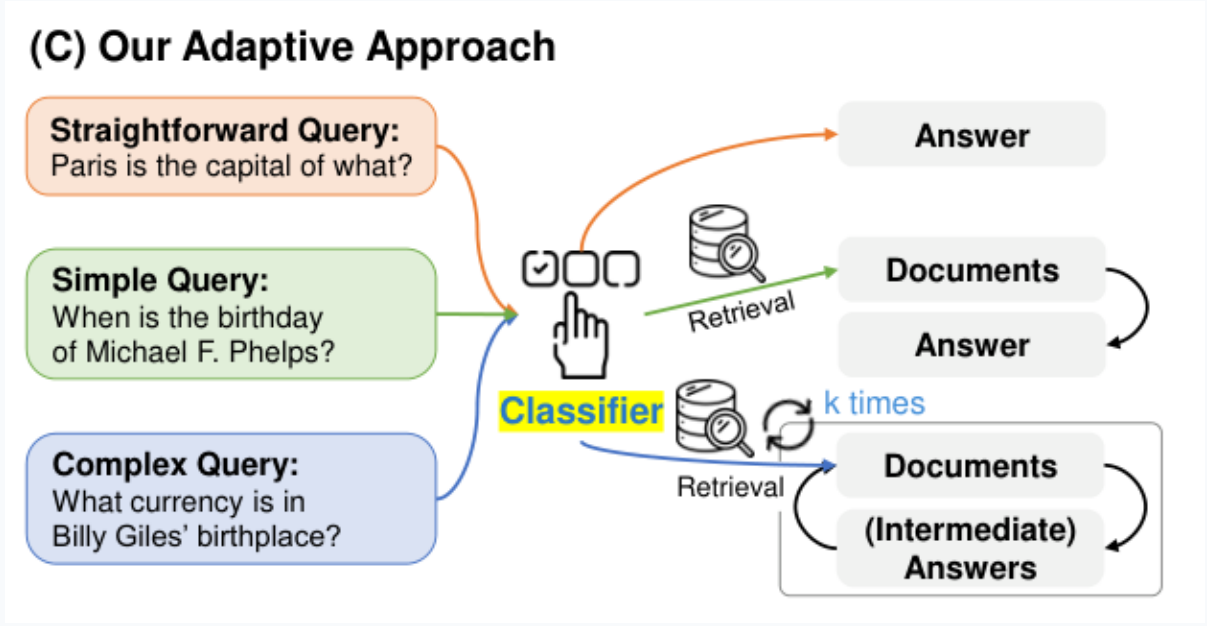


Single-step (A) and multi-step (B) approaches. Source: Jeong et al., NAACL 2024, Figure 2.

- **Simple:** "Paris is the capital of what?" (the model can answer alone)
- **Complex:** "When did the people who captured Malakoff come to the region where Philipsburg is located?" (4 reasoning steps)

- **Single-step search** is not enough for complex questions
- **Multi-step search** wastes time on simple questions

Adaptive-RAG: Match the Effort to the Question



The adaptive approach with a query-complexity classifier. Source: Jeong et al., NAACL 2024, Figure 2 (C).

A: answer without search

B: search once, then answer

C: search several times, reasoning between the steps

A small T5-Large classifier chooses A, B or C before any search

Training the Classifier without Manual Labels

STEP 1

Run all three strategies on training questions

STEP 2

Label = the simplest strategy that answers correctly

FALLBACK

No strategy is correct: label by dataset type (single-hop → B, multi-hop → C)

EXAMPLE

Example: an unlabelled HotpotQA question gets label C

Experimental Setup

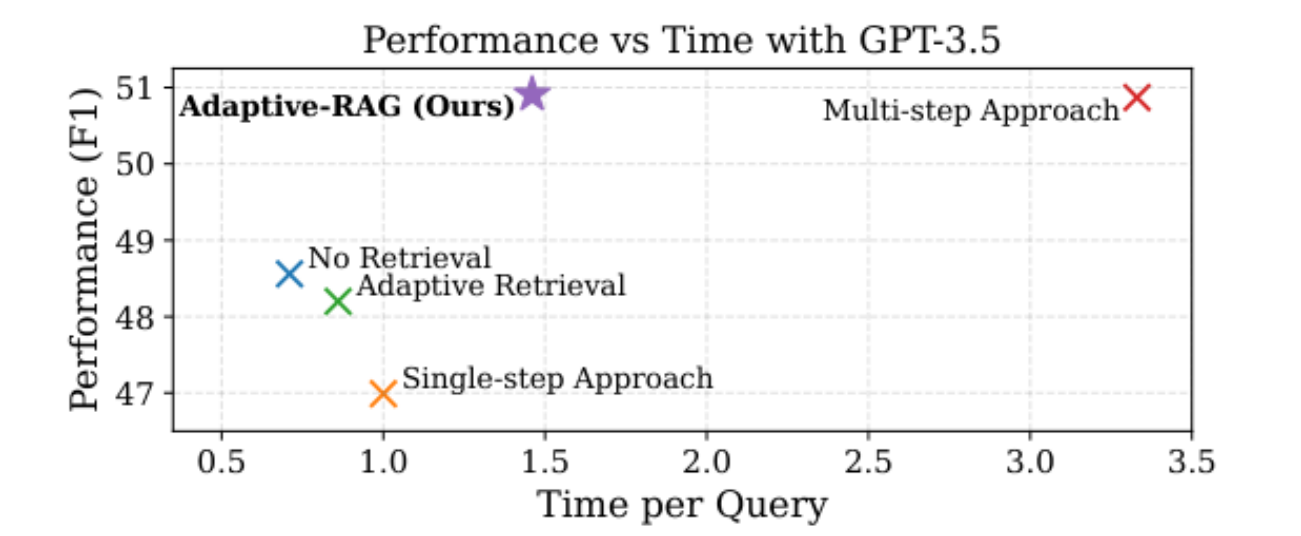
6 open-domain QA datasets

- **Single-hop:** SQuAD, Natural Questions, TriviaQA
- **Multi-hop:** MuSiQue, HotpotQA, 2WikiMultiHopQA

3 LLMs

- FLAN-T5-XL (3B)
- FLAN-T5-XXL (11B)
- GPT-3.5

Results with GPT-3.5



QA performance (F1) against time per query with GPT-3.5. Source: Jeong et al., NAACL 2024, Figure 1.

Adaptive-RAG: F1 50.91, relative time 1.46

Multi-step approach: F1 50.87, relative time 3.33

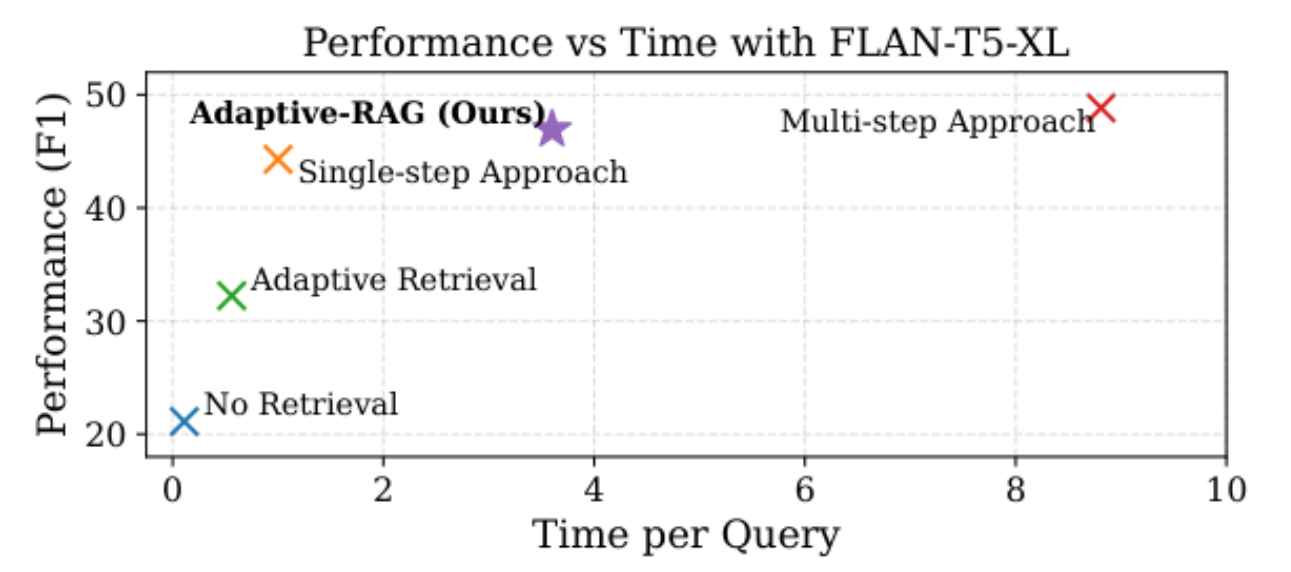
The same accuracy in less than half of the time

Results with FLAN-T5-XL (3B)

Adaptive-RAG: F1 46.94, relative time 3.60

Multi-step approach: F1 48.85, relative time 8.81

Adaptive Retrieval: F1 32.24, so Adaptive-RAG is 14.7 points higher



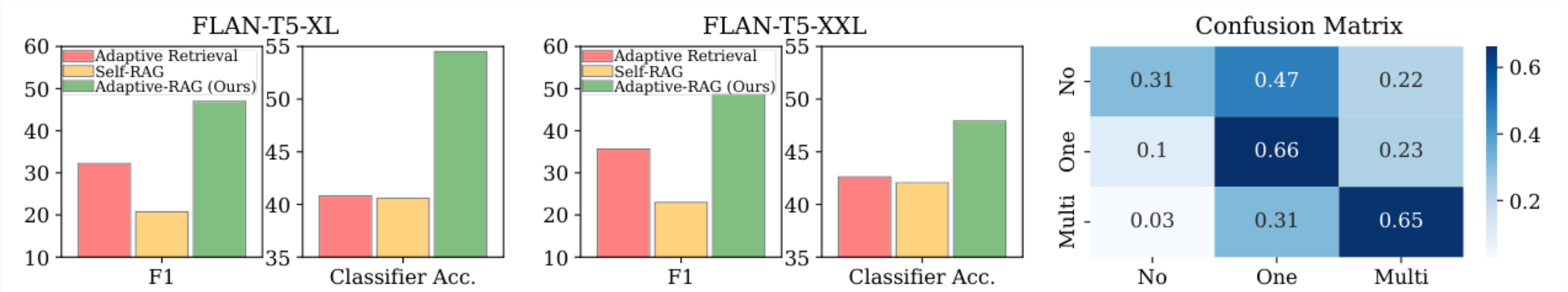
QA performance (F1) against time per query with FLAN-T5-XL. Source: Jeong et al., NAACL 2024, Figure 4.

Example: One Simple Question

Question: "Which famous corporate logo changed to a flat colour/color sans serif font in its first major change since 1999?"

Adaptive Retrieval	Adaptive-RAG
label B	label A
searched documents	no search
answer "Microsoft" (wrong)	**answer "Google" (correct)**

The Weak Point: The Classifier



Classifier performance with FLAN-T5-XL and XXL (left, centre) and the confusion matrix (right). Source: Jeong et al., NAACL 2024, Figure 3.

- Correct label for 54.5% of questions
- Easy questions are often sent to search
- With an ideal (oracle) classifier: F1 62.80 with GPT-3.5

Conclusions

1

Questions differ in complexity: one fixed strategy wastes resources or fails

2

Choosing a strategy per question: accuracy close to multi-step search in less than half of the time

3

Next steps: better complexity labels and better classifiers

KEY MESSAGE

"Before spending resources on an answer, ask how difficult the question really is"

Thank you! Questions?